

Born Robust, Then Calibrated

Continuous-attractor topology outlives the metric along it by an order of magnitude — how far depends on which developmental errors the circuit makes

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Abstract

Continuous attractor networks (CANs) underlie neural codes for head direction, grid and place. Their textbook weakness is the *fine-tuning problem*: an exact continuum of fixed points is a measure-zero object, so any asymmetry in the recurrent weights makes the bump drift. We argue this states the problem about the wrong object, and we separate it into two parts that behave completely differently. Using a ring CAN we show that (i) the *metric* along the manifold is what disorder destroys first, at first order and linearly: the formerly marginal Goldstone eigenvalue lifts off zero as $\lambda_{\parallel} \approx 1.9\epsilon$, with the leading drift rate given by the projection of the perturbation onto the Goldstone mode ($r = 0.99$, slope 0.98); (ii) the *topology* outlives it by an order of magnitude, the invariant circle and the localized bump surviving far past the loss of marginality; but (iii) how far depends on which errors the circuit actually makes. Under i.i.d. synaptic noise the traced loop remains resolved throughout the tested range, out to a perturbation comparable to the scaffold itself; under per-neuron heterogeneity — a structured error that moves the excitation/inhibition balance at $O(\eta)$ rather than $O(\eta/\sqrt{N})$ — the traced loop falls to the no-loop null at $\eta \approx 0.18$ while the bump is still coherent, and unlike the i.i.d. case that threshold does not improve with network size. (iv) The decomposition also has a computable validity boundary, where the on-manifold rate reaches the transverse contraction rate and normal hyperbolicity — with it Fenichel’s persistence guarantee — fails. (v) On the torus, where two Goldstone modes replace one, the traced manifold carries the required $b_1 = 2$ and disorder breaks the degeneracy of the pair, predicting unequal integration errors along the two directions of T^2 . We then show the fragile part is learnable. A three-factor, self-supervised weight rule, comparing an efference copy of self-motion against the velocity the recurrence produces on its own, learns the velocity kernel *from zero* and calibrates path-integration gain in the dark to $g = 0.99$ from a clean scaffold and $g = 1.02$ from a scaffold carrying substantial synaptic disorder ($\epsilon = 0.3$). The learned kernel is translation invariant (99.6% of its power) although nothing in the rule imposes symmetry, and its *shape* is not the one we hand-built but the derivative of the bump’s own autocorrelation ($\cos = 0.99$, versus 0.77 against the hand-built kernel). Both the symmetry and the shape of the learned connectivity are dictated by the statistics of the activity that trained it. The fine-tuning problem thus decomposes into a robust part that can be born and a low-dimensional fragile part that experience calibrates — with the robust part’s robustness now quantified, bounded, and conditional on the kind of developmental error the circuit is subject to. Finally, a Hebbian rule on undirected spontaneous activity builds the center-surround scaffold shape; after selecting a global gain near marginality it supports a ring attractor. The tested temporal-Hebbian rule requires a signed self-motion signal to form a velocity kernel, providing one mechanism compatible with topology preceding accurate integration.

1 Introduction

CANs explain the head-direction ring, the grid/place torus, the oculomotor line integrator and bump working memory [Khona and Fiete, 2022, Burak and Fiete, 2009]. The classical construction assumes translation-invariant connectivity $W(\mathbf{x}, \mathbf{y}) = f(\mathbf{x} - \mathbf{y})$; a continuum of marginally stable states then exists only on a measure-zero surface in parameter space, so small asymmetries destroy it [Ságodi et al., 2024, Amari, 1977]. This is the fine-tuning problem, and it is usually stated as a problem about the *existence* of the attractor.

The field has produced two largely separate responses. A *learning* thread grows the required connectivity by local plasticity — trace/Hebbian rules [Stringer et al., 2002], asymmetric STDP for grids [Widloski and Fiete, 2014], and a local self-supervised rule that learns the *Drosophila* ring connectome and attains gain-1 integration [Vafidis et al., 2022]. A *robustness* thread shows that imperfect attractors behave well enough: homeostatic plasticity restores the tuning condition [Gu and Lim, 2022], short-term plasticity pins the bump [Seeholzer et al., 2019], and continuous-symmetry dynamics survive disordered embeddings [Clark et al., 2025]. Ságodi et al. [2024] reframe the problem: what matters behaviourally is proximity to a *slow invariant manifold*, not exactness.

These threads collide with development. Head-direction attractor coupling is present at $\sim$ P12, before stable tuning and before eye opening [Bassett et al., 2018]; toroidal grid topology is reported before navigation [Guardamagna et al., 2026]. Yet accurate path integration — gain exactly 1 — demands an experience-dependent phase [Vafidis et al., 2022]. Something appears early and robustly; something else appears late and requires experience. No theory specifies which is which.

Claim. We claim the split is between topology and metric, and that it is a consequence of normal hyperbolicity rather than an assumption. A center-surround scaffold produces a slow manifold whose tangential rate is zero and whose transverse rates are $O(1)$. A small perturbation generally deforms such a manifold transversally by $O(\epsilon)$ and induces an $O(\epsilon)$ flow along it; normal hyperbolicity bounds the deformation and preserves the diffeomorphism type. So the ring survives — and with it the bump state, the population geometry, and everything an experimentalist measures as “topology” — while the marginality that makes the attractor *useful* is destroyed immediately and linearly. Topology can be born; gain must be calibrated.

We make this quantitative in a ring CAN, locate the boundary where the decomposition itself fails, show that a local rule repairs the fragile part, and verify that the calibrated network integrates a naturalistic head trajectory while a network with the same bump but a mis-set gain does not. We then ask how far the robust half actually goes, since structural stability is a statement about a *class* of perturbations and the class turns out to matter more than the thresholds do: i.i.d. synaptic noise, the standard caricature, is the benign case, while the resolved loop under heterogeneity across cells reaches the no-loop null at a spread of $\sim 18\%$ — an operational bound that, unlike the i.i.d. one, does not improve as the network grows. We carry the analysis to the torus, where two Goldstone modes replace one and disorder generically breaks their degeneracy, predicting path integration that is anisotropic in a way idiosyncratic to each animal. Finally we ask where the assumed scaffold comes from. A Hebbian rule on undirected spontaneous activity builds its center-surround shape; an oracle-selected global gain then places the learned kernel near marginality. The tested velocity rule needs a signed self-motion signal, motivating rather than proving the developmental order.

2 Model

We study a rate network of $N = 256$ neurons with preferred angles $\theta_i \in S^1$,

$$\tau \dot{\mathbf{r}} = -\mathbf{r} + \phi(W\mathbf{r} + \mathbf{I}(t)), \quad (1)$$

with ϕ a saturating sigmoid. The innate scaffold is a coarse center-surround kernel $W_{ij}^s = (J_e e^{-(\theta_i - \theta_j)^2 / 2\sigma^2} - J_i) / N$, which supports a localized bump and hence a manifold $\mathcal{M} = \{\mathbf{r}^*(\psi)\}$ of bump states. We use two models of imprecise wiring. The first is quenched Gaussian synaptic noise $\delta W = \epsilon \xi / \sqrt{N}$, $\xi_{ij} \sim \mathcal{N}(0, 1)$, the standard choice, with ϵ the single knob for “how sloppy the innate wiring is”. The second gives every neuron its own excitatory gain, inhibition and arbor width with fractional spread η (Methods). The distinction is not cosmetic — the first perturbs the excitation/inhibition balance at $O(\epsilon/\sqrt{N})$ and the second at $O(\eta)$ — and §3.5 shows it decides how robust the topology is. Velocity enters through an antisymmetric component, and the path-integration gain g is defined by $\dot{\psi} = g\omega$ under a self-motion signal $\omega(t)$.

Separating the two claims requires care. The natural measurement — relax bumps from many seed angles under a perturbed W and ask whether they stay put — does *not* measure topology. Fenichel’s theorem says the invariant circle survives; what it stops being is a circle of *fixed points*. It becomes a circle carrying a slow flow with isolated stable and unstable points, and free relaxation collapses every sample onto those few points. The measured “ring” then degrades even though the manifold is perfectly intact, and the measured drift is reported at exactly the locations where drift vanishes by construction. We therefore trace the manifold by *weak-pinning continuation*: hold the bump at a target angle with a small input, making it a fixed point of the augmented system, then release it briefly so the fast transverse directions relax onto $\mathcal{M}$ while the slow coordinate barely moves (Methods). Topology is then read off the traced sample by persistent homology, and the metric by the exact on-manifold flow $v(\psi) = \frac{d}{dt} \arg \sum_i r_i e^{i\theta_i}$.

The same care is needed in the opposite direction, and it becomes decisive once disorder is strong. A finite sample can fail to exhibit a loop that exists, so a *decline* in a persistence statistic is not by itself evidence of a topological change — it may only mean the manifold has become geometrically noisier. Every claim of lost topology below therefore carries a resolution control: refining the sample must recover a loop that is merely under-resolved and must fail to recover one that is genuinely gone (Methods). Where the control shows the statistic has saturated, we report the measurement as uninformative rather than as evidence of loss.

3 Results

3.1 The scaffold produces one marginal direction and a contracting bulk

The Jacobian of (1) at a bump state has a single eigenvalue at $\lambda = +0.002 \approx 0$ — the Goldstone mode of the broken translation symmetry — separated from the contracting bulk by a gap of 0.60 (Fig. 1a). This is the normal hyperbolicity condition: rate zero along $\mathcal{M}$, rate ≥ 0.6 transverse to it. Tracing the manifold recovers a clean circle whose persistent homology has a single H_1 bar of lifetime 0.86 (Fig. 1b), against a theoretical maximum of 0.87 for an ideal circle and a null of 0.07 (95th percentile 0.09) for a matched cloud with no loop.

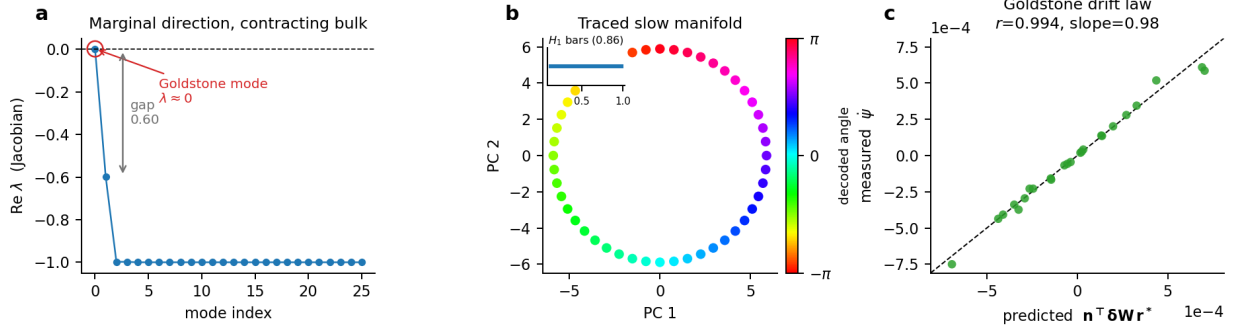


Figure 1: **The decomposition.** (a) Jacobian spectrum at a bump state: one marginal Goldstone eigenvalue, isolated from the contracting bulk by a gap of 0.60 — normal hyperbolicity. (b) The slow manifold traced by weak-pinning continuation, in its two leading principal components, coloured by decoded angle. Inset: H_1 persistence barcode, a single dominant bar. (c) The Goldstone drift law (2) against measured drift over a random perturbation ensemble.

3.2 Perturbations deform the manifold and drive it at the Goldstone projection

A perturbation δW generally produces both an $O(\|\delta W\|)$ normal displacement of $\mathcal{M}$ and an on-manifold flow. Normal hyperbolicity controls the former; the leading rate of the latter is the projection onto the left null vector $\mathbf{u}$ of the Jacobian,

$$v_{\text{drift}}(\psi) = \frac{\mathbf{u}(\psi)^\top D_\psi \delta W \mathbf{r}^*(\psi)}{\mathbf{u}(\psi)^\top \partial_\psi \mathbf{r}^*(\psi)} + O(\|\delta W\|^2), \quad (2)$$

where $D_\psi = \text{diag } \phi'$ gates the perturbation onto the unsaturated part of the bump (derivation in Methods). Over a random ensemble of perturbations, the measured drift matches (2) with correlation $r = 0.99$ and slope 0.98 (Fig. 1c). A perturbation orthogonal to the Goldstone mode produces no drift at first order, although it may still deform the manifold. **The fine-tuning-sensitive condition is the one-dimensional requirement $v_{\text{drift}} \equiv 0$, not the existence of $\mathcal{M}$.**

3.3 Under i.i.d. synaptic disorder, topology is structurally stable; marginality is not

Sweeping the disorder amplitude ϵ over eight seeds separates the two claims sharply (Fig. 2a,c). The bump state is untouched: coherence is flat at $0.854 \rightarrow 0.850$ across the entire range, so the localized activity pattern — the thing a recording would show — never degrades. The loop persists: there is *exactly one* H_1 bar at every disorder level tested, and at the largest disorder its persistence (0.31) is still $3.4\times$ the 95th percentile of a matched no-loop null (0.09). Its persistence does decline steadily from 0.86 (Fig. 2a), and we read that as the circle being less *cleanly* sampled rather than destroyed — the bar count and the null comparison, not the raw persistence, are what carry the claim.

The metric behaves entirely differently (Fig. 2c). The formerly marginal eigenvalue lifts off zero *linearly in the perturbation*, $\lambda_{\parallel} \approx 1.9\epsilon$ for $\epsilon \leq 0.2$, as first-order perturbation theory requires, while the transverse rate $\lambda_{\perp}$ stays flat near 0.6. Even a 2% perturbation converts a marginal direction into one with a rate of 0.03. Marginality is not robust to any degree; topology is robust to the end of the range.

3.4 The margin of the decomposition

The two rates approach each other (Fig. 2c). Clear timescale separation is already lost by $\epsilon_{1/2} \approx 0.17$, where $\lambda_{\parallel}/\lambda_{\perp}$ reaches $\frac{1}{2}$, and the ratio reaches 1 at $\delta_{\text{NH}} \approx 0.78$. We report both because the approach is gradual rather than a sharp transition: the ratio hovers near unity over $\epsilon \in [0.6, 1.5]$, and the interpolated unit-crossing varied between 0.67 and 1.06 across independent runs before we raised the seed count. The robust statement is that both thresholds are $O(1)$: the innate scaffold may be substantially sloppy — provided its sloppiness is of this kind, a qualification the next section makes precise and which turns out to matter more than the thresholds themselves.

Below δ_{NH} the manifold is normally hyperbolic, Fenichel’s theorem applies, and topology-versus-metric is a meaningful decomposition; above it the “slow” manifold is no longer slow relative to its transverse directions and the reduction to a scalar $\dot{\psi}$ loses its justification. Note that the bump and the loop both survive well past this point — *the theory’s own assumptions fail before the biology does*. This is the honest form of the “coarse-but-coherent window”: what bounds the tolerable disorder is not the destruction of the attractor but the loss of the timescale separation that makes the one-dimensional description — and the learning rule built on it — well posed.

3.5 How far structural stability goes depends on the disorder model

The i.i.d. model is the standard caricature of imprecise wiring, and it is also the benign one. The tell is in Fig. 2a: bump coherence is flat to three decimals across a $75\times$ range of ϵ . A perturbation $\epsilon\xi/\sqrt{N}$ moves the excitation/inhibition balance only at $O(\epsilon/\sqrt{N})$, so it can lift the Goldstone eigenvalue without ever threatening the bump that carries the manifold. Structural stability tested only against this model has not been tested hard.

Developmental error is more plausibly heterogeneity in cells and processes than independent noise on N^2 synapses. Giving each neuron its own excitatory gain, inhibition and arbor width with fractional spread η (Methods) perturbs the balance at $O(\eta)$, and the topology claim does not survive it (Fig. 2b). H_1 persistence falls to the no-loop null at $\eta \approx 0.18$ and to 0.03 — below the null — by $\eta = 0.30$.

This is not the bump dying. Coherence at $\eta = 0.30$ is still 0.79. What fails operationally is the tracer’s ability to place the bump *everywhere*: the decoded angles of the traced samples span $6.15 \rightarrow 5.41 \rightarrow 3.99$ radians at $\eta = 0, 0.20, 0.30$, and the residual between where the bump is pinned and where it settles grows from 0 to 0.65 rad. This is consistent with fragmentation into a shrinking set of allowed positions, but is not by itself proof that no invariant loop exists, especially because normal hyperbolicity has already failed. The loss of resolved H_1 is not a simple point-count artefact: tripling the trace density ($48 \rightarrow 160$ points) leaves it unchanged, whereas a genuinely under-resolved loop recovers under the same treatment (Methods).

Read against the metric, the decomposition’s qualitative claim survives. Under heterogeneity marginality is half-gone by $\eta_{1/2} = 0.012$ and normal hyperbolicity fails at $\eta = 0.08$, while the resolved loop persists to $\eta \approx 0.18$ — still an order of magnitude of separation between the two failures. Two things change. First, the measured robustness is now *finite*, and it lands inside the biologically plausible range: an 18% spread in cellular parameters is not obviously beyond what a developing circuit exhibits, so “born robust” is a claim about a developmental program that controls those parameters, not a free lunch. Second, the two models are not interchangeable even at matched metric damage (Fig. 2d): at $\lambda_{\parallel}/\lambda_{\perp} \approx 1.12$, i.i.d. noise leaves $H_1 = 0.31$ and heterogeneity leaves 0.08. I.i.d. disorder cannot push the system far enough past the normal-hyperbolicity boundary to destroy the loop; heterogeneity can.

Scaling with N . The two models also scale differently, which decides how much of this is an artefact of the size we happened to simulate (Table 1). Tolerable i.i.d. disorder *grows* strongly with network size: $\epsilon_{1/2}$ more than triples from $N = 128$ to $N = 512$ and δ_{NH} leaves the tested range entirely, while the slope $\lambda_{\parallel}/\epsilon$ stays nearly constant. That is what a $1/\sqrt{N}$ -normalized perturbation acting on a fixed number of relevant Fourier modes should do, and it means the $O(1)$ i.i.d. bounds quoted above are conservative for biological N .

Heterogeneity does not behave this way. Over the same four-fold range, $\eta_{1/2}$ only doubles and the amplitude at which the loop is lost is flat within noise (0.19, 0.18, 0.23). The asymmetry has a simple origin: i.i.d. noise is normalized by N and is diluted as the network grows, whereas heterogeneity is a fixed *fractional* error on each neuron’s kernel and is not. The consequence is the one that matters biologically — robustness to cell-level heterogeneity cannot be bought by adding neurons, and the $\sim 18\%$ threshold is a property of the circuit’s parameters rather than of $N = 256$. (These runs use a coarser amplitude grid and fewer seeds than the headline sweeps, so absolute values sit slightly off those quoted above; the trends are the result.)

N	i.i.d. (ϵ)			heterogeneous (η)	
	$\epsilon_{1/2}$	δ_{NH}	$\lambda_{\parallel}/\epsilon$	$\eta_{1/2}$	η at loop loss
128	0.080	0.40	1.67	0.008	0.187
256	0.190	0.58	1.64	0.013	0.181
512	0.283	> 1.2	1.38	0.016	0.232

Table 1: Tolerance to i.i.d. noise grows with network size; tolerance to cell-level heterogeneity does not.

3.6 A local rule learns the velocity kernel and calibrates the gain

If only the metric is fragile, plasticity has a narrow and well-posed job. We implement the self-supervised rule at the level of the weights themselves, with *no velocity kernel supplied*: the network holds two plastic matrices,

$$W(t) = W^s + \Delta(t) + \omega(t) M(t), \quad (3)$$

driven by the same error and the same local factors,

$$\dot{M} = \eta_M e(t) \omega(t) \frac{(\partial_{\psi} \mathbf{r}) \mathbf{r}^{\top}}{N} - \gamma M, \quad \dot{\Delta} = \eta_{\Delta} e(t) \frac{(\partial_{\psi} \mathbf{r}) \mathbf{r}^{\top}}{N} - \gamma \Delta. \quad (4)$$

The implementation is a three-factor abstraction rather than a fully specified synaptic circuit. The error compares an efference copy of self-motion against the velocity the recurrence produces *on its own*, $e(t) = \omega(t) - \dot{\psi}_{\text{rec}}(t)$. In code, $\dot{\psi}_{\text{rec}}$ is a population-vector decoder and $\partial_{\psi} \mathbf{r}$ is a finite difference across neighbouring preferred angles. Thus no target angle is supplied, but an ordered coordinate and circuit mechanisms for these two non-synapse-local factors are assumed. Since $\dot{\psi}_{\text{rec}} = g\omega + v_{\text{drift}}$, the error decomposes as $e = (1 - g)\omega - v_{\text{drift}}$: multiplying by ω extracts the gain error and trains M , while the part uncorrelated with ω reports drift and trains Δ . One rule, two gating conditions, two jobs.

Starting from $M = 0$, the gain measured *in the dark* (recurrence only, no sensory input — a condition never trained on directly) rises to $g = 0.99$ on the clean scaffold and $g = 1.02$ on a scaffold carrying substantial synaptic disorder ($\epsilon = 0.3$), with the mean error falling five-fold (Fig. 3a). That the disordered case calibrates just as well is the point: the network does not need to know its own wiring, only to observe what that wiring does. We stress what this does and does not show. The

disordered scaffold is the designed kernel plus quenched noise, not connectivity generated from statistics alone in the sense of [Clark et al. \[2025\]](#); the ring coordinate is still supplied. What is demonstrated is that calibration tolerates a badly specified scaffold, not that the scaffold can be dispensed with.

3.7 Both the symmetry and the shape of the learned kernel come from activity statistics

Nothing in (4) refers to $\theta_i - \theta_j$, yet the learned M is translation invariant: decomposed into its circulant part and the remainder, the circulant part carries 99.6% of its power on the clean scaffold and 80.5% on the disordered one, and its profile is odd to a correlation of 0.998.

The mechanism is where the symmetry that the classical construction *assumes* actually comes from. The outer product $(\partial_\psi \mathbf{r}) \mathbf{r}^\top$ is localized wherever the bump currently is, and is not translation invariant at any instant. But its expectation over a bump that explores the ring is

$$\mathbb{E}_\psi [(\partial_\psi \mathbf{r}^*(\psi)) \mathbf{r}^*(\psi)^\top]_{ij} = -\partial_\Delta C(\Delta)|_{\Delta=\theta_i-\theta_j}, \quad C(\Delta) = \int r^*(\theta) r^*(\theta + \Delta) d\theta, \quad (5)$$

a function of $\theta_i - \theta_j$ alone, and odd: the derivative of the bump’s autocorrelation. The symmetry of the learned weights is inherited from the ergodicity of the bump’s motion.

Equation (5) makes a sharper prediction than symmetry, and the simulation confirms it. The learned kernel does *not* match the velocity kernel we hand-built — a Gaussian derivative, cosine similarity 0.77. It matches $-\partial_\Delta C$ to cosine similarity 0.99 (Fig. 3b). Because this network’s transfer function saturates, its bump is nearly a boxcar, its autocorrelation nearly a triangle, and the learned kernel accordingly a broad, flat-topped odd square wave rather than a narrow Gaussian derivative — 55% of its entries lie within 90% of its maximum, against 10% for the hand-built kernel. It is a different velocity kernel from the designed one, and it integrates just as accurately. What the connectivity looks like is set by the statistics of the activity that trained it, not by the modeller.

This also supplies the identifiability condition: if the animal does not move, the gain is unobservable and nothing is learned; if it moves but does not explore, the kernel is learned only locally. In our runs the head traverses ~ 8 laps of the ring, which suffices.

3.8 Drift repair is partial

The static term Δ addresses residual drift. Its error signal is unambiguous only when little self-motion is commanded — otherwise it is swamped by the gain error — and the positions needing repair are precisely those a freely relaxing bump never visits. Running it as a landmark-anchored phase, with the bump held at swept angles and the recurrent drift nulled at each, reduces RMS on-manifold drift by $3.3\times$ (Fig. 3c). We report this as a partial result: the reduction is real and the sign and magnitude of the correction were verified directly, but convergence across passes is noisy and we do not claim the clean exponential repair that the gain calibration achieves.

3.9 The calibrated network path-integrates a naturalistic trajectory

Gain in the dark is a scalar; the behaviour it underwrites is the accurate integration of a real head trajectory, which is how ring-attractor function is usually demonstrated [[Vafidis et al., 2022](#)]. We drive the network through its learned kernel with a naturalistic angular velocity $\omega(t)$ — an Ornstein–Uhlenbeck process, ~ 2 turns over the run — and decode the bump position against the true integrated heading $\int \omega dt$ (Fig. 4).

The comparison is the paper’s thesis rendered behavioural, because it isolates the metric from the topology. With the calibrated kernel the bump tracks the true heading with a slope-0.99 gain and a residual error of 3.6° RMS over the whole trajectory, the error staying bounded and diffusive (Fig. 4b, d). Scaling the *same* kernel to gain 0.7 — a network with the identical bump and identical topology but a mis-set metric — leaves a bump that still moves smoothly and still looks like a head-direction cell, yet whose heading estimate slips systematically behind the truth, accumulating to 111° RMS. Topology gives an integrator; only the calibrated metric makes it *accurate*. This is why the fragile, one-dimensional part of the connectivity is the part that has to be learned: a plausible developmental error in the gain does not abolish the head-direction representation, it silently corrupts the quantity the animal navigates by.

3.10 On the torus, disorder splits the two Goldstone modes

Grid cells make the toroidal case the one that touches population data [Guardamagna et al., 2026], and the extension is not automatic: T^2 carries *two* Goldstone modes, so disorder need not lift them equally. Anisotropic lifting has no one-dimensional analogue, and it is the substantive new observable here.

The ring parameters do not transfer. Excitatory mass under the bump scales as σ^2 rather than σ , and the same neuron budget spread over two dimensions leaves a bump only a few lattice sites across, which locks to the lattice — the discretization, not the disorder, then breaks the continuous symmetry. Because that failure mode would counterfeit precisely the effect we are looking for, we measure it directly. For the *clean* scaffold the residual $|\lambda_{\parallel}|/\text{gap}$ falls $6.7\% \rightarrow 2.4\% \rightarrow 1.4\% \rightarrow 0.4\%$ as the lattice refines from 24^2 to 48^2 , while the overlap between the leading two-dimensional eigenspace and the span of the two translation directions rises $0.74 \rightarrow 0.96$ (Fig. 5a). The slow pair is the translation pair, and its residual is a finite-lattice artefact rather than a property of the model.

At 40^2 ($N = 1600$) the traced manifold has the homology T^2 requires (Fig. 5b,c). The top two H_1 bars are equal — as the scaffold’s $x \leftrightarrow y$ symmetry demands — at $L_1 = L_2 = 0.605$ against a third bar of 0.116, a two-bar margin of 5.2. Reference clouds at matched sampling put that in context: an ideal square torus scores 4.3, a sphere 1.1, a Gaussian blob 1.0. They also show why two statistics are needed rather than one, since the sphere has near-equal top bars too ($L_2/L_1 = 0.88$) and would pass a symmetry test on its own. The two Goldstone eigenvalues are degenerate at $\lambda_1 = \lambda_2 = 0.0092$ against $\lambda_{\perp} = 0.69$, and the drift vanishes.

Under disorder the metric fails as it does on the ring: the slow rates rise from 0.009 to 0.30 while $\lambda_{\perp}$ holds near 0.7, and the bump is untouched across the whole i.i.d. range (coherence $0.844 \rightarrow 0.820$). The two-loop signature degrades more gradually, from margin 5.2 to 2.4 at $\epsilon = 0.3$, still well clear of the sphere and blob controls (1.1, 1.0). Beyond $\epsilon \approx 0.5$ we stop drawing conclusions, because the resolution calibration says we cannot: a *genuine* square torus carrying 20% geometric noise also scores 1.1, so the margins of 1.2–1.7 measured at large disorder are equally consistent with an intact-but-noisy torus and a destroyed one. We report the second loop as confirmed to $\epsilon \approx 0.3$ and unresolved beyond, not as lost. Heterogeneity again does the same damage some fifteen times cheaper, the margin leaving the resolved range by $\eta \approx 0.05$ rather than $\epsilon \approx 0.5$. At $\eta \geq 0.20$ the margin appears to *recover*, but this is an artefact worth naming: the bump itself is disintegrating there (coherence $0.84 \rightarrow 0.48$, longest bar $0.61 \rightarrow 0.09$), so L_2/L_3 becomes a ratio of vanishing numbers. Those points are drawn hollow in Fig. 5c and carry no topological information.

The splitting is the observable the ring cannot provide. On the production grid, whose trace points are commensurate with the neural lattice, the clean pair is exactly degenerate by symmetry;

any disorder breaks it, and the measured anisotropy $(\lambda_1 - \lambda_2)/(\lambda_1 + \lambda_2)$ jumps immediately to 0.4–0.56 and stays there under both models, with no trend in amplitude (Fig. 5d). That is precisely the magnitude random-matrix theory predicts for a generically perturbed degenerate pair — between 0.33 and 0.47 according to whether the effective 2×2 block is taken symmetric or general (Methods) — so the splitting is not anomalous. It is unavoidable. Nothing in an isotropic scaffold perturbed by isotropic noise preserves the degeneracy, and the consequence is a testable prediction rather than a demonstrated computation: if the split slow rates map onto unequal integration errors, a toroidal attractor built this way should path-integrate more accurately along one direction of T^2 than the other, with the preferred axis set by that particular network’s quenched wiring error rather than by anything in its design or its environment.

3.11 The scaffold can be built from spontaneous activity

Everything so far *assumes* W^s . The obvious objection is that a tuned Mexican hat is exactly the fine-tuned object the paper set out to avoid needing. We close by asking whether the scaffold can instead be built, and find that the same fault line that runs through the rest of the paper — topology cheap, metric dear — also governs its construction.

Suppose neurons carry a coordinate (θ_i) and that spontaneous activity is smooth, localized, and drifts around the ring, exploring it (a caricature of the propagating activity present in these circuits before experience). We generate such activity as a rough bump — with multiplicative per-sample noise, *not* a clean template — whose centre executes an Ornstein–Uhlenbeck walk, and apply the textbook plasticity rules (Methods). The symmetric scaffold is the fixed point of a Hebbian rule, and that fixed point is the row-centred activity covariance; subtractive normalization turns co-activation into a centre and anti-correlation into a surround, so a Mexican hat appears with no inhibition put in by hand (Fig. 6a; cosine 0.88 with the hand-built kernel, the residual being a difference in width set by the activity’s correlation length).

After scanning a single global scale and selecting the point closest to marginality, the learned kernel supports a ring attractor rather than merely a bump former. Its Goldstone eigenvalue sits at -0.009 with a spectral gap of 0.99 to the contracting bulk, its traced manifold is a single H_1 loop, and — the check that the ring is encoded in the connectivity rather than only in the labels we supplied — its two leading eigenvectors are the $\cos \theta, \sin \theta$ Fourier modes, overlap 1.00 (Fig. 6b). None of this used a self-motion signal: undirected spontaneous activity is enough to build the topology. This is the mechanistic content of “topology can be born”.

Two caveats keep this honest. First, marginality is exact only in the *ensemble*: because no ring position is special a priori, the expected Hebbian increment is translation-invariant, but a finite amount of activity leaves an $O(1/\sqrt{\text{events}})$ non-circulant residual — itself quenched disorder, which pins the bump exactly as in §3.5. That residual falls from 0.16 to 0.03 as the activity grows from ~ 10 to ~ 70 laps of the ring (Fig. 6c): *more spontaneous experience yields a cleaner, less pinned scaffold*, and the very quantity that limits it is the metric disorder the rest of the paper studies. Second, and by contrast, the velocity kernel is not identifiable under the tested ungated temporal-Hebbian rule. Its circulant part is proportional to the *mean* rotation $\langle v \rangle$ of the activity; for undirected spontaneous activity ($\langle v \rangle = 0$) it vanishes (amplitude $< 10^{-4}$), and only gating the rule by a signed self-motion signal — an efference copy, proportional to $\langle v^2 \rangle$ — produces a kernel ($56\times$ larger, Fig. 6d) of the right shape. For this rule and an unbiased motion ensemble, a signed gating signal is required; other developmental rules or directionally biased waves are not excluded.

These results provide one mechanism compatible with the developmental order — topology before accurate integration — rather than establishing that order as a necessary consequence. What remains assumed, and is the honest boundary, is the coordinate itself: “localized in θ ” and

“drifts around the ring” presuppose the ordering. This derives the scaffold’s shape and, after global gain selection, a correctly-labelled ring attractor from a locality prior plus activity; it does not yet derive the ring from unlabelled neurons.

4 Discussion

The fine-tuning problem has been stated about the existence of the attractor. Our results say existence is the robust part. What disorder destroys, immediately and linearly, is marginality — and marginality is a property of the *flow on* the manifold, a one-dimensional quantity, not of the manifold’s existence.

This has four consequences. First, it dissolves an apparent paradox in the developmental data: ring and toroidal topology can appear before experience [Bassett et al., 2018, Guardamagna et al., 2026] because topology holds on an open set of connectivities while marginality holds on a measure-zero one. Accurate path integration appears later [Vafidis et al., 2022] because gain is the measure-zero condition, and calibrating it requires the very self-motion statistics that early behaviour provides. This ordering is not merely consistent with the decomposition but mechanistically forced by it (§3.11): the scaffold is the fixed point of a Hebbian rule that needs only undirected spontaneous activity, whereas the velocity kernel is the fixed point of a rule whose signal vanishes unless it is gated by an efference copy of self-motion. Topology can be learned before behaviour because its learning rule does not reference behaviour; the gain cannot.

The heterogeneity results say how large that open set is, and the answer is not “arbitrarily”. Topology tolerates roughly an order of magnitude more perturbation than marginality does under either disorder model, but under cell-level heterogeneity it fails at a spread of $\sim 18\%$, and that bound does not soften as the circuit grows. This turns a qualitative reassurance into a quantitative developmental requirement: a circuit can be sloppy about individual *synapses* and still be born with the right manifold — i.i.d. errors of order the scaffold itself leave the loop intact, and the tolerance improves with N — but it must control the gains, arbor widths and inhibitory tone of individual *cells* to within roughly a fifth of their means, or possess homeostatic machinery that does so. Which of the two error types dominates in a given system is measurable, and it decides whether “born robust” is cheap or expensive there.

Second, it sharpens what plasticity must do. Learning models that grow the whole attractor from scratch [Stringer et al., 2002, Widloski and Fiete, 2014] are solving a harder problem than biology needs to. Assigning plasticity the narrower job of metric calibration is also what makes the rule identifiable and its convergence clean, as Fig. 3a shows.

Third, it identifies where the structure of the connectivity comes from. The classical construction assumes translation invariance; Clark et al. [2025] and Mastrogiuseppe and Ostojic [2018] show continuous-symmetry dynamics can arise from connectivity *statistics*. We add a mechanism on the learning side, and it delivers more than symmetry. A rule with no spatial structure produces a translation-invariant kernel because the bump samples the ring — the symmetry of the weights is the symmetry of the behaviour that trained them — and (5) then fixes the kernel’s *shape* as the derivative of the bump’s autocorrelation. The network did not recover the kernel we designed; it recovered the one its own activity implies, and integrated just as accurately with it. This suggests a testable reading of measured velocity-tuned connectivity: its profile should track the autocorrelation of the local activity pattern rather than any canonical functional form.

Fourth, the toroidal case adds an observable the ring cannot express. Where a ring has one gain to calibrate, a torus has two, and disorder does not damage them equally: the degeneracy of the Goldstone pair is generically broken. Anisotropic path integration is then not evidence of anything

special about an animal’s environment or anatomy — it is the expected signature of quenched wiring error, with an axis that should vary from animal to animal. It also poses a harder problem for plasticity than the one we solved here. The rule of (4) calibrates a single scalar gain along a single Goldstone direction; on a torus whose two modes disorder has already separated, that reduction is not available, and whether one local rule can calibrate both gains without cross-talk is open.

Predictions. The decomposition is falsifiable. (i) Manifold topology (persistent homology of population activity) should be measurable before behavioural gain reaches 1. (ii) Early-experience manipulations should perturb gain and anchoring while leaving topology intact. (iii) Because calibration requires self-motion statistics, restricting movement during the critical window should impair gain specifically, leaving the manifold. (iv) If the same rule continuously re-tunes, blocking plasticity in the adult should cause slow gain drift; if the attractor is built-then-frozen, adult blockade is inert. This last experiment distinguishes “continuously re-tuned” from “built-then-maintained”. (v) Because disorder breaks the degeneracy of the two toroidal Goldstone modes, grid-cell path integration should be measurably *anisotropic* on the torus — drift along one axis of the toroidal coordinate exceeding the other — and, because the splitting comes from that animal’s own quenched wiring error, the preferred axis should be idiosyncratic across animals rather than aligned to any external or anatomical direction. This is the sharpest of the five: an ensemble statistic that is nonzero in every animal but has no population-level mean direction is hard to produce any other way. (vi) Manipulations that increase cell-to-cell parameter variability should degrade manifold topology far more effectively per unit of measured synaptic noise than manipulations that add independent synaptic noise, and increasing the number of neurons should not rescue the former.

Limitations. The most important is the coordinate. Section 3.11 removes the assumption that the *scaffold* is hand-built — it emerges from spontaneous activity — but it still hands neurons a preferred angle θ_i : “localized in θ ” and “drifts around the ring” presuppose the ordering, and the emergence result recovers the Fourier coordinate rather than constructing the ring from unlabelled cells. So the present work derives the scaffold’s *shape* and its marginality from a locality prior plus activity, but not the ring’s existence from scratch. Building the coordinate itself — from developmental gradients, or from connectivity statistics in the sense of Clark et al. [2025], and then recovering the topology by an unsupervised embedding rather than by the supplied labels — is the step that would fully earn the first half of our title. On T^2 this is sharper still: the two-dimensional coordinate and the commensurability of the two periods are both assumed.

Second, learning is shown on the ring only. The torus results cover the decomposition, not calibration, and the two-mode case poses a question the ring cannot: whether one rule can calibrate two gains that disorder has already made unequal, given that the modes it must separate are no longer degenerate.

Third, the heterogeneity model, though far more damaging than i.i.d. noise, is still a caricature — independent jitter of three parameters per neuron, with no spatial correlation in the developmental error and no compensating homeostasis. Real circuits have both, and both would move the threshold, plausibly in opposite directions.

Finally, the learning implementation is not yet a fully local biological rule: $e(t)$ is a population-level modulatory quantity, ψ is computed by a population-vector decoder, and $\partial_\psi r_i$ uses neighbouring preferred-angle units. Circuit realizations of these factors remain predictions rather than results; the dense signed rate network also does not enforce Dale’s law, sparse connectivity, conductance dynamics, or spiking noise. Drift repair is partial (§3.8).

5 Methods

Network. $N = 256$, $\tau = 1$, $J_e = 60$, $J_i = 18$, $\sigma = 0.5$; $\phi(x) = (1 + e^{-(x-b)/w})^{-1}$ with $b = 0.5$, $w = 0.12$. Euler integration at $dt = 0.1$.

Disorder models. *i.i.d.*: $\delta W = \epsilon \xi / \sqrt{N}$, $\xi_{ij} \sim \mathcal{N}(0, 1)$, whose spectral norm is $\approx 2\epsilon$ independently of N , so ϵ is comparable across network sizes. *Heterogeneous*: each neuron i receives its own kernel parameters $J_e^{(i)} = J_e(1 + \eta z_1)$, $J_i^{(i)} = J_i(1 + \eta z_2)$, $\sigma^{(i)} = \sigma e^{\eta z_3}$ with $z_k \sim \mathcal{N}(0, 1)$, and $W_{ij} = (J_e^{(i)} e^{-d_{ij}^2/2\sigma^{(i)2}} - J_i^{(i)})/N$. This is a $3N$ -parameter, spatially structured perturbation that moves the excitation/inhibition balance at $O(\eta)$, whereas i.i.d. noise moves it at $O(\epsilon/\sqrt{N})$.

Torus network. $n_s \times n_s$ preferred phases on T^2 , isotropic kernel $W_{ij}^s = (J_e e^{-d_{ij}^2/2\sigma^2} - J_i)/N$ with $d_{ij}^2 = \Delta x_{ij}^2 + \Delta y_{ij}^2$ wrapped to $(-\pi, \pi]$. The 1-D parameters do not transfer: excitatory mass under the bump scales as σ^2 rather than σ , and a bump only a few lattice sites across locks to the lattice. Scanning gave $J_e = 60$, $J_i = 23$, $\sigma = 1.0$ ($\sim 12\%$ of the population in the bump); $\sigma \geq 1.2$ delocalizes it. Production runs use $n_s = 40$ ($N = 1600$). Manifold tracing is as on the ring over a 10×10 grid of target centers, with hold 900 steps.

Torus topology. $b_1(T^2) = 2$, so the signature is a *second* long H_1 bar, not a long one. From the top three bar lifetimes $L_1 \geq L_2 \geq L_3$ we report L_2/L_3 (topological: does a second loop stand above the short-bar noise floor?) and L_2/L_1 (geometric: are the two loops the same size?). These must not be conflated — a torus with one small loop is still a torus. Both are calibrated against reference clouds of known homology at matched sampling (`torus.validate_score`).

Manifold tracing. For each of 48 target angles: form a bump (300 steps), hold with input $0.10 \mathbf{b}(c)$ for 1200 steps, release for 40 steps. The release time is several transverse relaxation times ($1/0.6$) but short enough that the slow coordinate moves negligibly.

Topology. Vietoris–Rips persistent homology over $\mathbb{Z}/2$ computed from scratch (`code/topology.py`), on the traced states projected to their 6 leading principal components, distances normalised by cloud diameter. The reported statistic is the longest H_1 bar. Null: matched isotropic Gaussian clouds of the same size and dimension.

Distinguishing loss of a resolved loop from undersampling. A finite point sample can fail to show a loop that exists, so a fall in H_1 is not by itself evidence of a topological change — the same conflation of topology with geometry that motivates §2. The control is sampling density: an under-resolved loop can recover when the sample is refined. Failure to recover is evidence against a simple point-count artefact, but does not prove nonexistence of an invariant loop. Refining the traced ring from 48 to 96 to 160 points leaves the heterogeneous case flat ($H_1 = 0.021, 0.021, 0.015$ at $\eta = 0.15$), while the second bar of a deliberately under-resolved reference torus recovers over the same refinement (L_2/L_3 $1.07 \rightarrow 1.59$ from 64 to 100 points). We also observed that even-stride subsampling of a grid-sampled torus aliases against the grid period and moves the margin non-monotonically ($2.69/1.03/1.95$ at $100/80/64$ points on one cloud), so torus homology is always computed on the full sample.

Rates. Both rates come from the Jacobian spectrum along the traced manifold: $\lambda_{\parallel}$ is the largest real part (the perturbed Goldstone eigenvalue), $\lambda_{\perp}$ the second largest. An earlier estimator computed $\lambda_{\parallel}$ as $\max |dv/d\psi|$ from the traced flow; that is the same quantity in principle but a poor estimator, since differentiating samples that carry pinning jitter amplifies it without bound (the estimate grew by three orders of magnitude with fit resolution, without converging). The eigenvalue requires no differentiation.

Drift law. $\mathbf{u}$ is the left null vector of the Jacobian at the unperturbed bump; $\partial_{\psi}\mathbf{r}^*$ a centred difference along the ring. Measured drift is the slope of the unwrapped decoded angle. Ensemble of 24 perturbations at $\epsilon = 0.015$.

Learning. $\omega(t)$ is Ornstein–Uhlenbeck ($\sigma_{\omega} = 0.05$, $\tau_v = 100$), giving ~ 8 laps of the ring over 4×10^4 steps — enough exploration for the kernel to become translation invariant. A landmark input of amplitude 1.0 drags the bump; it does not enter the error. $\eta_M = 100$, $\gamma = 10^{-6}$. Gain probes run with the sensory input removed. Note the sign convention: $\partial_{\psi}\mathbf{r} = -\partial_{\theta}\mathbf{r}$ because $r_i = f(\theta_i - \psi)$; the opposite sign turns (4) into positive feedback and the weights diverge.

Path integration (§3.9). A naturalistic angular velocity $\omega(t)$ is drawn as an Ornstein–Uhlenbeck process ($\sigma = 0.06$, $\tau = 40$, $dt = 0.1$) over 6000 steps and fed to the network through the learned kernel, $W = W^s + g\omega(t)M$; the bump is decoded at every step and compared to the true heading $\int \omega dt$. The gain is the slope of the per-step decoded rotation on ωdt ; the under-calibrated condition simply sets the scalar $g = 0.7$ on the identical kernel. The reported RMSE and Fig. 4d use the cumulative unwrapped heading error; wrapped circular errors are also cached.

Torus rates and the anisotropy null. The two slow rates and the transverse rate are read from the Jacobian spectrum at sampled points of the traced manifold, as on the ring. The anisotropy is computed *per state* and then averaged: taking the maximum over states of each eigenvalue separately and comparing the two maxima biases the splitting downward, because each rate attains its maximum at a different point of the manifold. The null is the anisotropy expected when the degenerate pair is perturbed by a random 2×2 block. Two ensembles bracket it and we quote both: GOE if the block is symmetric (mean 0.47), real Ginibre if general (mean 0.33, lower because 29% of its draws have a complex-conjugate pair whose real parts are exactly equal). The Jacobian is not normal, so the truth lies between. An incommensurate tracing grid can itself create position-dependent splitting of the clean pair; production uses $n_s = 40$ and a commensurate 10×10 trace, and the smoke run uses $n_s = 32$ with an 8×8 trace. This is a discretization control, not evidence that finite lattices have exact continuous symmetry.

Scaffold from activity (§3.11). Spontaneous activity: a bump of width 0.6 rad with 35% multiplicative per-sample noise, its centre following an Ornstein–Uhlenbeck angular walk ($\sigma_v = 0.1$, $\tau_v = 60$) that covers the ring; “undirected” means the realized $\langle v \rangle$ is set to zero. Symmetric scaffold: W^s = the time-covariance of the activity with each row centred (subtractive normalization), diagonal removed, then rescaled by a single global gain selected by an oracle scan to minimize the Goldstone rate while retaining a coherent bump and spectral gap. A dynamical homeostatic rule that finds this operating point is not implemented. The reported kernel is the circulant (ensemble) part, with the non-circulant residual quoted separately. Velocity kernel: the antisymmetric temporal correlation $\sum_t g_t [\mathbf{r}(t)\mathbf{r}(t-1)^{\top} - \mathbf{r}(t-1)\mathbf{r}(t)^{\top}]$, ungated ($g_t = 1$) or gated by the signed

self-motion signal ($g_t = v_t$); the reported amplitude is the peak of its odd circulant profile. Coordinate recovery: mean squared cosine between the span of the two leading eigenvectors of the learned W and the span of $\cos \theta, \sin \theta$.

Reproduction. `python run_paper.py` regenerates the ring numbers (~ 8 min), `python run_torus.py` the torus numbers (~ 12 min), `python het_scaling.py` Table 1 (~ 3 min), `python run_formation.py` Fig. 6 (~ 20 s) and `python run_pathint.py` Fig. 4 (~ 20 s); `python figures.py` redraws all six figures from the caches. Set `OMP_NUM_THREADS` to a modest value: the inner loop is a bandwidth-bound matrix–vector product, and unbounded BLAS threading slowed the torus runs by more than an order of magnitude.

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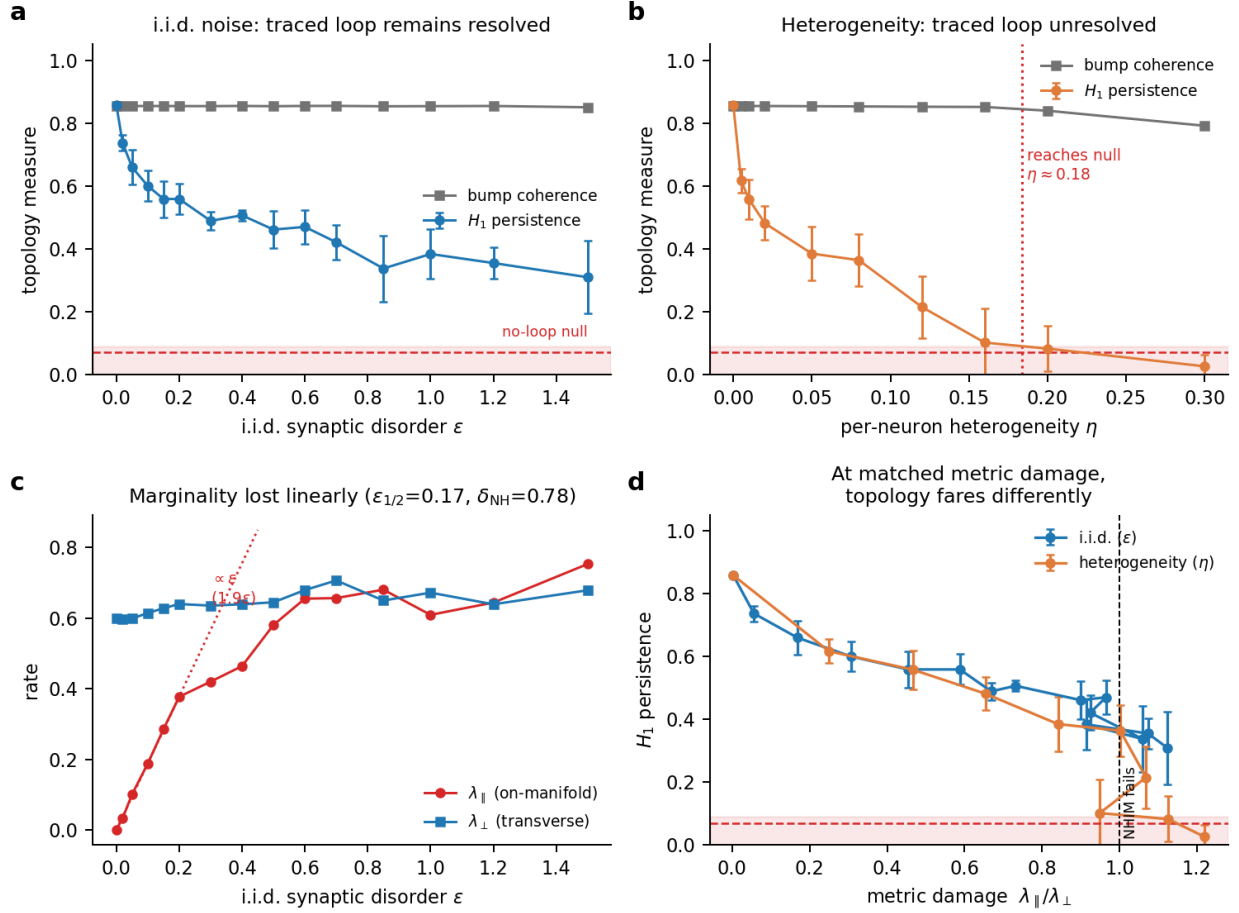


Figure 2: **Topology is structurally stable under i.i.d. synaptic noise, but not under cell-level heterogeneity.** Mean $\pm$ s.d. over 8 disorder seeds; red band is the 95th percentile of a matched no-loop null. **(a)** i.i.d. disorder: bump coherence is flat and H_1 persistence stays well above the null out to $\epsilon = 1.5$. **(b)** Per-neuron kernel heterogeneity: H_1 falls to the null at $\eta \approx 0.18$, while the bump is still coherent (0.79 at $\eta = 0.30$). **(c)** The on-manifold rate $\lambda_{\parallel}$ grows linearly from zero, $\approx 1.9\epsilon$, while $\lambda_{\perp}$ stays flat near 0.6; clear timescale separation is lost by $\epsilon_{1/2} \approx 0.17$ and $\lambda_{\parallel}/\lambda_{\perp}$ reaches 1 near $\delta_{NH} \approx 0.78$. **(d)** Both models plotted against the metric damage each has already done, the only common axis available since ϵ and η are not the same units. The curves track while $\lambda_{\parallel}/\lambda_{\perp} < 1$ and separate past it: at matched damage ≈ 1.12 , i.i.d. noise leaves $H_1 = 0.31$ while heterogeneity leaves 0.08, below the null.

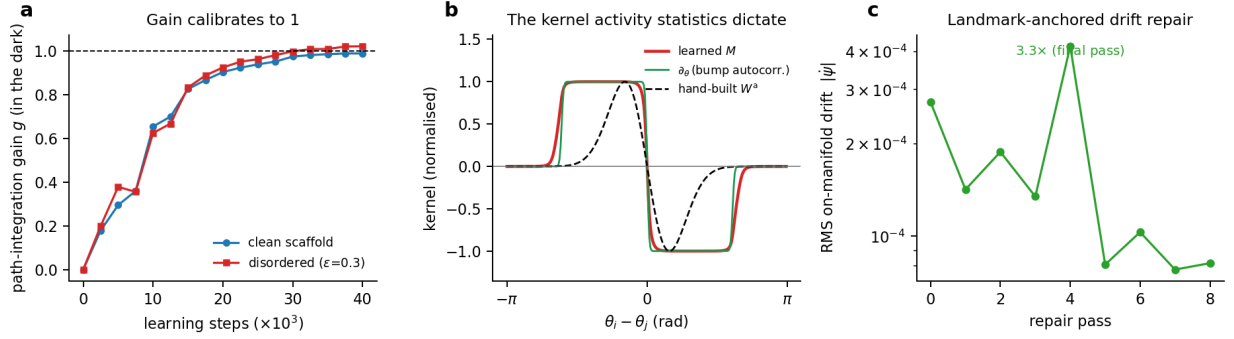


Figure 3: **The fragile part is learnable.** (a) Path-integration gain in the dark during learning, from a clean and from a disordered ($\epsilon = 0.3$) scaffold, both converging to 1 from $M = 0$. (b) The learned kernel M , averaged into its circulant profile, against the derivative of the bump’s autocorrelation (5) (cosine similarity 0.99) and against the hand-built velocity kernel (0.77). The rule never refers to $\theta_i - \theta_j$: both the translation invariance and the shape are inherited from the activity that trained it. (c) Landmark-anchored drift repair; the trace is noisy across passes and the quoted factor is the final pass, not the best one.

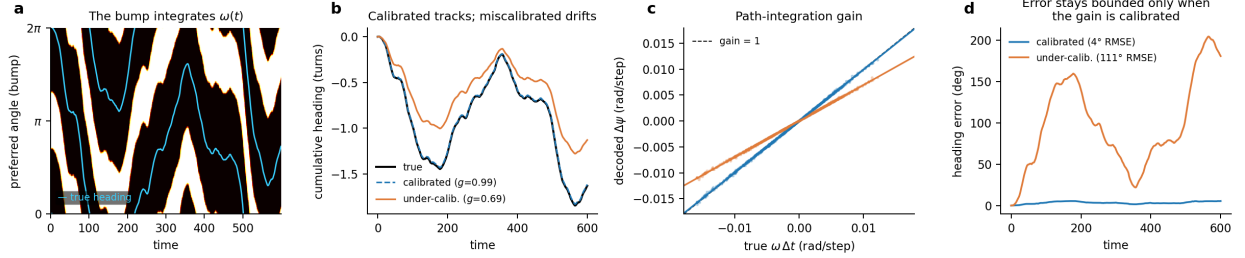


Figure 4: **Path integration of a naturalistic head-direction trajectory.** (a) Population activity (bump) driven by $\omega(t)$; the cyan curve is the true heading, which the bump follows. (b) Decoded cumulative heading against the truth: the calibrated kernel ($g = 0.99$) overlies the true trace, an under-calibrated kernel ($g = 0.69$) drifts away. (c) Per-step decoded rotation versus the true angular-velocity step; the fitted slope is the path-integration gain. (d) Accumulated heading error. It stays bounded (3.6° RMS) only when the gain is calibrated; a mis-set gain grows to 111° on the same bump.

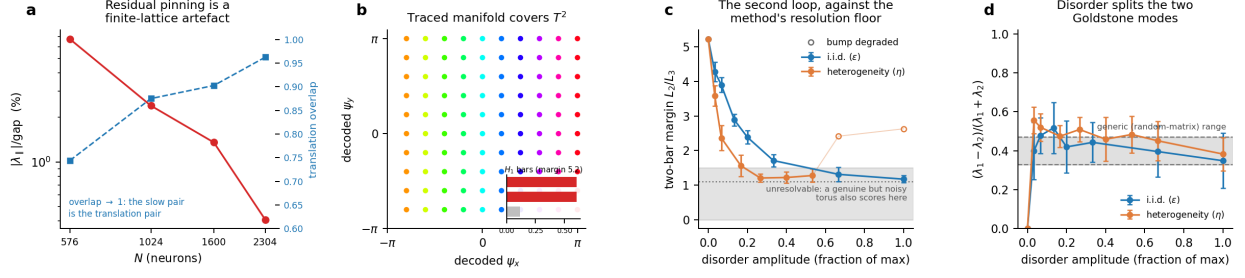


Figure 5: **The decomposition on T^2 .** (a) Residual pinning of the clean scaffold against lattice size (red, left axis) and the overlap of the leading two-dimensional eigenspace with the two translation directions (blue, right). The marginality is genuine, not a discretization artefact. (b) Decoded positions of the traced samples cover the torus. Inset: the top three H_1 bar lifetimes — two long, one short — which is the T^2 signature. (c) The two-bar margin under both disorder models, against the shaded range in which the statistic cannot distinguish a lost second loop from a merely noisy one (a genuine torus with 20% geometric noise scores 1.1, dotted). Hollow markers are points where bump coherence has fallen below $0.9\times$ its clean value, so the ratio is formed from vanishing bar lifetimes and means nothing. (d) The degeneracy of the Goldstone pair is broken by any disorder, by the amount expected for a generic perturbation. Grey band spans the two random-matrix nulls (Ginibre 0.33, GOE 0.47).

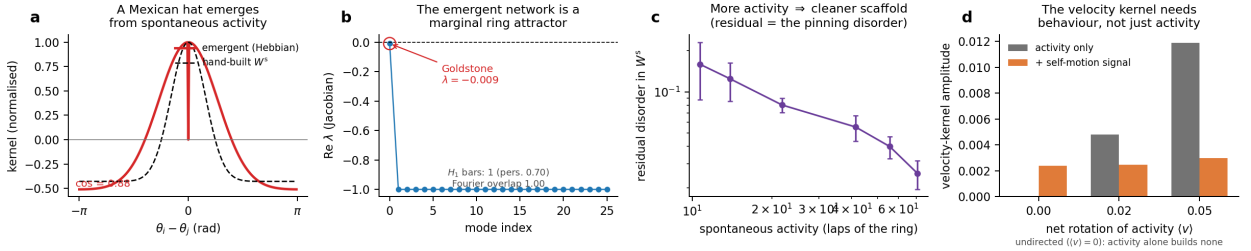


Figure 6: **A scaffold shape is learned from spontaneous activity.** (a) The circulant profile of the Hebbian-learned kernel (red) against the hand-built Mexican hat (dashed); a centre-surround kernel emerges with no inhibition specified. (b) After selecting a global gain, the Jacobian spectrum of the learned kernel has a single near-zero Goldstone mode over a contracting bulk. Its manifold is one H_1 loop and its leading eigenvectors are the Fourier modes (overlap 1.00) — a marginal ring attractor, built from undirected activity. (c) The non-circulant residual disorder of the learned scaffold falls with the amount of spontaneous activity; this residual is what pins the bump. (d) Velocity-kernel amplitude versus the net rotation $\langle v \rangle$ of the activity. The activity-only (ungated) rule tracks $\langle v \rangle$ and gives nothing for undirected activity; only the self-motion-gated rule builds a kernel.